🚀 LECTURE 02 : OPERATOR , CONDITIONAL , SWITCH , BREAK , CONTINUE

📘 1. Operators

➕ A. Arithmetic Operators

Used for math calculations.

```
int a = 10, b = 3;

cout << a + b;   // 13 (addition)
cout << a - b;   // 7  (subtraction)
cout << a * b;   // 30 (multiplication)
cout << a / b;   // 3  (division)
cout << a % b;   // 1  (remainder/modulo)
```

✅ Example

```
int marks1 = 80, marks2 = 90;

int total = marks1 + marks2;
int average = total / 2;

cout << "Total: " << total << endl;       // 170
cout << "Average: " << average << endl;   // 85
```

📝 B. Assignment Operators

Used to assign values.

```
int x = 10;

x = 5;      // x becomes 5
x += 3;     // x = x + 3  → x becomes 8
x -= 2;     // x = x - 2  → x becomes 6
x *= 4;     // x = x * 4  → x becomes 24
x /= 2;     // x = x / 2  → x becomes 12
```

⚖️ C. Comparison Operators

Used to compare values. Result is `true` or `false`.

```cpp
int a = 10, b = 5;

a == b    // false (is equal?)
a != b    // true  (is not equal?)
a > b     // true  (is greater?)
a < b     // false (is smaller?)
a >= b    // true  (is greater or equal?)
a <= b    // false (is smaller or equal?)
```

🔗 D. Logical Operators

Combine multiple conditions.

```cpp
// AND (&&) - Both must be true
true && true     // true
true && false    // false

// OR (||) - At least one must be true
true || false    // true
false || false   // false

// NOT (!) - Reverses
!true     // false
!false    // true
```

✅ Example

```cpp
int age = 20;
bool hasLicense = true;

// Can drive if age >= 18 AND has license

if(age >= 18 && hasLicense) {
    cout << "Can drive!";
}
```

🔼 E. Increment / Decrement

Quick way to add or subtract 1.

```cpp
int x = 5;
```

```
x++;   // x becomes 6
x--;   // x becomes 5
```

✅ Example

```
int score = 0;

score++;  // score becomes 1
score++;  // score becomes 2
score++;  // score becomes 3
```

⚡ F. Bitwise Operators

Work directly on binary bits (`0` and `1`).

```
int a = 5, b = 3;

// AND (&)
cout << (a & b);   // 1

// OR (|)
cout << (a | b);   // 7

// XOR (^)
cout << (a ^ b);   // 6

// NOT (~)
cout << (~a);      // -6
```

📌 Common Bitwise Operators

```
a & b   // AND
a | b   // OR
a ^ b   // XOR
~a      // NOT
```

✅ Example

```
5 = 0101
3 = 0011
```

```
5 & 3 = 0001  // 1
5 | 3 = 0111  // 7
5 ^ 3 = 0110  // 6
```

📌 Mostly Used In

- Competitive Programming
- Fast calculations
- Binary operations
- DSA & System Programming

↔ G. Bitwise Shift Operators

Used to shift bits left or right.

```
int x = 5;

// Left Shift (<<)
cout << (x << 1);   // 10
cout << (x << 2);   // 20

// Right Shift (>>)
cout << (x >> 1);   // 2
cout << (x >> 2);   // 1
```

⬅ Left Shift `<<`

Moves bits to the left.

👉 Each left shift multiplies the number by 2.

```
5 << 1   // 10
5 << 2   // 20
5 << 3   // 40
```

📌 Binary Example

```
5 = 00000101

5 << 1

00000101 → 00001010
```

✅ Result

```
10
```

➡️ Right Shift `>>`

Moves bits to the right.

👉 Each right shift divides the number by 2.

```
20 >> 1   // 10
20 >> 2   // 5
20 >> 3   // 2
```

📌 Binary Example

```
5 = 00000101

5 >> 1

00000101 → 00000010
```

✅ Result

```
2
```

✅ Example

```
int salary = 1000;

cout << (salary << 1) << endl; // 2000
cout << (salary >> 1) << endl; // 500
```

📌 Shortcut Formula

```
x << n    // x × (2^n)
x >> n    // x ÷ (2^n)
```

⌨️ 2. Input and Output

🖥️ Output (`cout`)

Print to screen.

```
cout << "Hello World";
cout << "Age: " << 25;
cout << "Name: " << name << endl;
```

⌨️ Input (`cin`)

Take input from user.

```
int age;

cout << "Enter your age: ";
cin >> age;

string name;

cout << "Enter your name: ";
cin >> name;
```

✅ Complete Example

```
#include <iostream>
using namespace std;

int main() {

    string name;
    int age;

    cout << "Enter your name: ";
    cin >> name;
```

```
    cout << "Enter your age: ";
    cin >> age;

    cout << "\nHello " << name << "!" << endl;
    cout << "You are " << age << " years old." << endl;

    return 0;
}
```

🔀 3. If-Else Statements

✅ Basic If Statement

```
if(condition) {
    // code runs if condition is true
}
```

✅ Example

```
int age = 20;

if(age >= 18) {
    cout << "You are an adult";
}
```

🔁 If-Else Statement

```
if(condition) {
    // code runs if condition is true
} else {
    // code runs if condition is false
}
```

✅ Example

```
int age = 15;

if(age >= 18) {
    cout << "You are an adult";
} else {
```

```
    cout << "You are a minor";
}
```

🔄 If-Else If-Else Statement

```
if(condition1) {
    // runs if condition1 is true
} else if(condition2) {
    // runs if condition2 is true
} else {
    // runs if all conditions are false
}
```

✅ Example

```
int marks = 85;

if(marks >= 90) {
    cout << "Grade: A";
} else if(marks >= 80) {
    cout << "Grade: B";
} else if(marks >= 70) {
    cout << "Grade: C";
} else if(marks >= 60) {
    cout << "Grade: D";
} else {
    cout << "Grade: F";
}
```

🧩 Nested If (If inside If)

```
int age = 20;
bool hasLicense = true;

if(age >= 18) {

    if(hasLicense) {
        cout << "You can drive";
    } else {
        cout << "You need a license";
    }
```

```
} else {
    cout << "You are too young";
}
```

🧪 Practice Problems

✅ Problem 1 : Even or Odd

```
#include <iostream>
using namespace std;

int main() {

    int num;

    cout << "Enter a number: ";
    cin >> num;

    if(num % 2 == 0) {
        cout << num << " is even";
    } else {
        cout << num << " is odd";
    }

    return 0;
}
```

✅ Problem 2 : Maximum of Two Numbers

```
#include <iostream>
using namespace std;

int main() {

    int a, b;

    cout << "Enter first number: ";
    cin >> a;

    cout << "Enter second number: ";
    cin >> b;

    if(a > b) {
        cout << a << " is maximum";
    } else {
```

```
        cout << b << " is maximum";
    }

    return 0;
}
```

✅ Problem 3 : Positive, Negative, or Zero

```
#include <iostream>
using namespace std;

int main() {

    int num;

    cout << "Enter a number: ";
    cin >> num;

    if(num > 0) {
        cout << "Positive";
    } else if(num < 0) {
        cout << "Negative";
    } else {
        cout << "Zero";
    }

    return 0;
}
```

✅ Problem 4 : Calculator

```
#include <iostream>
using namespace std;

int main() {

    double num1, num2;
    char op;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter operator (+, -, *, /): ";
    cin >> op;
```

```
    cout << "Enter second number: ";
    cin >> num2;

    if(op == '+') {
        cout << "Result: " << (num1 + num2);
    } else if(op == '-') {
        cout << "Result: " << (num1 - num2);
    } else if(op == '*') {
        cout << "Result: " << (num1 * num2);
    } else if(op == '/') {

        if(num2 != 0) {
            cout << "Result: " << (num1 / num2);
        } else {
            cout << "Cannot divide by zero!";
        }

    } else {
        cout << "Invalid operator!";
    }

    return 0;
}
```

✅ Problem 5 : Grade Calculator

```
#include <iostream>
using namespace std;

int main() {

    int marks;

    cout << "Enter your marks (0-100): ";
    cin >> marks;

    if(marks > 100 || marks < 0) {
        cout << "Invalid marks!";
    } else if(marks >= 90) {
        cout << "Grade: A (Excellent)";
    } else if(marks >= 80) {
        cout << "Grade: B (Very Good)";
    } else if(marks >= 70) {
        cout << "Grade: C (Good)";
    } else if(marks >= 60) {
        cout << "Grade: D (Satisfactory)";
    } else {
        cout << "Grade: F (Fail)";
    }
```

```
    return 0;
```

📌 Summary Cheat Sheet

📦 Variables

```
int age = 25;
double price = 99.99;
char grade = 'A';
bool flag = true;
string name = "Rohit";
```

⚡ Operators

```
+  -  *  /  %          // Math
=  +=  -=  *=  /=      // Assignment
==  !=  >  <  >=  <= // Comparison
&&  ||  !              // Logical
++  --                 // Increment/Decrement
```

⌨️ Input / Output

```
cout << "Text";        // Output
cin >> variable;       // Input
getline(cin, str);     // Input with spaces
```

🔀 If-Else

```
if(condition) {
    // code
} else if(condition) {
    // code
} else {
```

```
    // code
```

🔄 Switch Statement

❓ What is Switch?

Alternative to multiple `if-else if` statements.

Better for checking one variable against many values.

📌 Syntax

```
switch(variable) {

    case value1:
        // code
        break;

    case value2:
        // code
        break;

    default:
        // code if no match
}
```

✅ Example 1 : Days of Week

```
int day = 3;

switch(day) {

    case 1:
        cout << "Monday" << endl;
        break;

    case 2:
        cout << "Tuesday" << endl;
        break;

    case 3:
        cout << "Wednesday" << endl;
        break;
```

```
    case 4:
        cout << "Thursday" << endl;
        break;

    case 5:
        cout << "Friday" << endl;
        break;

    case 6:
        cout << "Saturday" << endl;
        break;

    case 7:
        cout << "Sunday" << endl;
        break;

    default:
        cout << "Invalid day" << endl;
}
```

✅ Output

```
Wednesday
```

✅ Example 2 : Calculator

```
char op;
double a, b;

cout << "Enter operator (+, -, *, /): ";
cin >> op;

cout << "Enter two numbers: ";
cin >> a >> b;

switch(op) {

    case '+':
        cout << "Result: " << (a + b) << endl;
        break;

    case '-':
        cout << "Result: " << (a - b) << endl;
        break;

    case '*':
        cout << "Result: " << (a * b) << endl;
        break;
```

```cpp
    case '/':

        if(b != 0) {
            cout << "Result: " << (a / b) << endl;
        } else {
            cout << "Cannot divide by zero!" << endl;
        }

        break;

    default:
        cout << "Invalid operator" << endl;
}
```

⚠️ Important : `break` Statement

Without `break`, code falls through.

```cpp
int num = 2;

switch(num) {

    case 1:
        cout << "One" << endl;

    case 2:
        cout << "Two" << endl;

    case 3:
        cout << "Three" << endl;

    default:
        cout << "Other" << endl;
}
```

✅ Output

```
Two
Three
Other
```

✅ With `break`

```
int num = 2;

switch(num) {

    case 1:
        cout << "One" << endl;
        break;

    case 2:
        cout << "Two" << endl;
        break;

    case 3:
        cout << "Three" << endl;
        break;

    default:
        cout << "Other" << endl;
}
```

✅ Output

```
Two
```

🔁 Continue Statement

`continue` skips the current iteration and moves to the next iteration of the loop.

✅ Example

```
for(int i = 1; i <= 5; i++) {

    if(i == 3) {
        continue;
    }

    cout << i << " ";
}
```

✅ Output

1 2 4 5